Building Information Modeling (BIM) Lab

Course Code: 23A01607 | L-T-P-C: 0-1-2-2 | B.Tech Civil Engineering | JNTUA R23

Experiment-wise Inputs and Outputs

The table below summarises, for each experiment in the syllabus, what the student provides or uses (inputs) and what is produced (outputs). Inputs and outputs are derived from the listed experiment topics.

No.	Experiment	Inputs	Outputs
1	**Introduction to BIM and Autodesk Revit**	• Revit software (project template: Architectural/Structural) • Basic BIM concepts and Revit terminology (element, category, family, type, instance) • Mouse/keyboard commands for viewing (zoom, pan, orbit)	• New project file (.rvt) created from a template • Familiarity with Ribbon, Project Browser, Properties palette, View Control Bar • Saved project with different view navigation demonstrated
2	**Basic Drawing and Editing Tools**	• Sketch dimensions for lines, rectangles, circles, arcs • Revit drawing tools (Line, Rectangle, Circle, Arc) • Modify tools: Move, Copy, Rotate, Mirror, Trim/Extend, Offset, Array, Align	• Simple 2D geometry drawn accurately • Edited elements using modification tools • Saved practice file (.rvt)
3	**Setting Up Levels and Grids**	• Number of storeys and floor-to-floor height • Grid spacing (column-to-column distances) in X and Y directions • Structural column sizes and types • A 2D CAD file (.dwg) to link or import	• Levels with elevations and associated plan views • Grid layout with bubbles and labels • Structural columns placed at grid intersections • Linked/imported CAD file positioned in the model
4	**Modeling Walls**	• Floor plan/layout (hand sketch or linked CAD) • Wall type, thickness, height and base/top constraints • Location line and material/finish choices	• Exterior wall shell enclosing the building • Interior partition walls placed and joined • Modified walls (edited profile, attached top/base, wall joins) • 3D view of the walled model
5	**Working with Doors and Windows**	• Door/window schedule: type, width, height, sill height • Family files (.rfa) from the Revit library • Host walls with defined openings	• Doors and windows inserted in walls at the correct positions • New loaded families and new type sizes created • Door/window tags or schedule (optional)
6	**Working with Curtain Walls**	• Curtain wall location and dimensions • Grid spacing (horizontal and vertical) • Panel types (glass, solid, door panel) and mullion profiles	• Curtain wall with a defined grid • Panels replaced/customised as required • Mullions attached to the grid lines • 3D and elevation views showing the curtain wall
7	**Working with Views**	• Completed building model • Required scale, detail level and visual style • Locations for sections, elevations and callouts	• Duplicated views (with detailing/dependent options) • Callout views of selected areas • Elevations (N, S, E, W) and sections of the model • View settings controlled (scale, detail level, visibility/graphics)

No.	Experiment	Inputs	Outputs
8	**Adding Components**	• Component families (furniture, fixtures, equipment) from the library • Placement locations and orientation	• Components placed in the model • Components modified (type change, move, rotate, copy, parameters) • Updated plan and 3D views
9	**Modeling Floors**	• Floor boundary (room outline or wall picks) • Floor type, thickness and level • Shaft opening size/location; slope direction and gradient	• Floors modelled at each level • Floor and wall geometry joined • Shaft openings created through floors • Sloped floor created (using slope arrow or edited points)
10	**Modeling Ceilings and Roofs**	• Ceiling height and ceiling type • Light fixture families • Roof type, slope/pitch, overhang and footprint	• Ceiling with fixtures placed • Ceiling soffit created • Roof model (by footprint/extrusion) attached to walls • 3D view and ceiling plan
11	**Modeling Stairs and Railings**	• Floor-to-floor height (to calculate risers and treads) • Stair width, tread depth, riser height and location • Railing type and ramp slope	• Component stair with correct runs, landings and railings • Modified stair (edited boundary, number of risers) • Custom sketched stair • Ramp created with railings • Completed 3D model

Experiment Details

1. INTRODUCTION TO BIM AND AUTODESK REVIT

Topics: About Autodesk and AutoCAD, workflow and BIM, Revit terms, overview of the interface, starting projects, viewing commands.
Inputs: Revit software (project template: Architectural/Structural); Basic BIM concepts and Revit terminology (element, category, family, type, instance); Mouse/keyboard commands for viewing (zoom, pan, orbit).
Outputs: New project file (.rvt) created from a template; Familiarity with Ribbon, Project Browser, Properties palette, View Control Bar; Saved project with different view navigation demonstrated.

2. BASIC DRAWING AND EDITING TOOLS

Topics: General drawing tools, editing elements, working with modification tools.
Inputs: Sketch dimensions for lines, rectangles, circles, arcs; Revit drawing tools (Line, Rectangle, Circle, Arc); Modify tools: Move, Copy, Rotate, Mirror, Trim/Extend, Offset, Array, Align.
Outputs: Simple 2D geometry drawn accurately; Edited elements using modification tools; Saved practice file (.rvt).

3. SETTING UP LEVELS AND GRIDS

Topics: Setting up levels and grids, creating structural grids, adding columns, linking and importing CAD files.
Inputs: Number of storeys and floor-to-floor height; Grid spacing (column-to-column distances) in X and Y directions; Structural column sizes and types; A 2D CAD file (.dwg) to link or import.
Outputs: Levels with elevations and associated plan views; Grid layout with bubbles and labels; Structural columns placed at grid intersections; Linked/imported CAD file positioned in the model.

4. MODELING WALLS

Topics: Modeling walls, modifying walls, modeling the exterior shell, adding interior walls.
Inputs: Floor plan/layout (hand sketch or linked CAD); Wall type, thickness, height and base/top constraints; Location line and material/finish choices.
Outputs: Exterior wall shell enclosing the building; Interior partition walls placed and joined; Modified walls (edited profile, attached top/base, wall joins); 3D view of the walled model.

5. WORKING WITH DOORS AND WINDOWS

Topics: Inserting doors and windows, loading door and window types from the library, creating additional sizes.
Inputs: Door/window schedule: type, width, height, sill height; Family files (.rfa) from the Revit library; Host walls with defined openings.
Outputs: Doors and windows inserted in walls at the correct positions; New loaded families and new type sizes created; Door/window tags or schedule (optional).

6. WORKING WITH CURTAIN WALLS

Topics: Creating curtain walls, adding curtain grids, working with curtain wall panels, attaching mullions.
Inputs: Curtain wall location and dimensions; Grid spacing (horizontal and vertical); Panel types (glass, solid, door panel) and mullion profiles.
Outputs: Curtain wall with a defined grid; Panels replaced/customised as required; Mullions attached to the grid lines; 3D and elevation views showing the curtain wall.

7. WORKING WITH VIEWS

Topics: Setting the view display, duplicating views, adding callout views, elevations and sections.
Inputs: Completed building model; Required scale, detail level and visual style; Locations for sections, elevations and callouts.
Outputs: Duplicated views (with detailing/dependent options); Callout views of selected areas; Elevations (N, S, E, W) and sections of the model; View settings controlled (scale, detail level, visibility/graphics).

8. ADDING COMPONENTS

Topics: Adding components, modifying components, working with elements.
Inputs: Component families (furniture, fixtures, equipment) from the library; Placement locations and orientation.
Outputs: Components placed in the model; Components modified (type change, move, rotate, copy, parameters); Updated plan and 3D views.

9. MODELING FLOORS

Topics: Modeling and modifying floors, joining geometry, creating shaft openings, creating sloped floors.
Inputs: Floor boundary (room outline or wall picks); Floor type, thickness and level; Shaft opening size/location; slope direction and gradient.
Outputs: Floors modelled at each level; Floor and wall geometry joined; Shaft openings created through floors; Sloped floor created (using slope arrow or edited points).

10. MODELING CEILINGS AND ROOFS

Topics: Modeling ceilings, adding ceiling fixtures, creating ceiling soffits, modeling roofs.
Inputs: Ceiling height and ceiling type; Light fixture families; Roof type, slope/pitch, overhang and footprint.
Outputs: Ceiling with fixtures placed; Ceiling soffit created; Roof model (by footprint/extrusion) attached to walls; 3D view and ceiling plan.

11. MODELING STAIRS AND RAILINGS

Topics: Creating component stairs, modifying component stairs, working with railings, sketching custom stairs, creating ramps.
Inputs: Floor-to-floor height (to calculate risers and treads); Stair width, tread depth, riser height and location; Railing type and ramp slope.
Outputs: Component stair with correct runs, landings and railings; Modified stair (edited boundary, number of risers); Custom sketched stair; Ramp created with railings; Completed 3D model.

Overall Course Output

By the end of the lab, the student produces a complete 3D Revit building model (levels, grids, columns, walls, doors, windows, curtain walls, floors, ceilings, roofs, stairs and railings) along with plans, elevations, sections and callout views, meeting Course Outcomes CO1 to CO5.